Sajan Toor

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Education

Simon Fraser University — *Burnaby, BC*
Bachelor's of Science: Computer Science — *Sept 2020 - Aug 2025*

Work Experience

SAP — *Jan 2025 - Present*
Software Engineer — *Vancouver, BC*

- Developed cloud-native microservices handling **millions of authentication, authorization, and routing requests**, using Java, Spring Boot, Resilience4j, Jakarta, Feign, and Project Reactor, powering SAP core infrastructure
- Designed a end-to-end testing CI/CD pipeline with multi-agent orchestration to detect regressions by analyzing test results, metrics, logs and traces, enabling AI agents and developers to contribute code with more confidence
- Built a full stack bundled authentication feature, enabling single identity provider (IdP) across multiple SAP products, simplified IdP federation and identity management, powered by OpenID Connect
- Improved performance of IdP switching workflows by **50x** by optimizing database queries and batching user updates
- Created a browser extension that analyzes HAR and SAML files, enabling support teams to **self-resolve 90%** of authentication issues, significantly reducing developer on-call load, **cutting resolution time by over 50%**
- Led the team in the number of production bugs resolved over the year
- Prevented long-term data growth and infrastructure costs by fixing a data orphaning issue early in production
- **Reduced request latency by 70%** through asynchronous and non-blocking operations using Project Reactor
- Saved 1 hour per developer per sprint across 12 teams by automating release branching, versioning, and changelogs

SAP — *May 2024 - Jan 2025*
Software Engineer Intern — *Vancouver, BC*

- Achieved **60% faster API responses** using Caffeine cache, boosting throughput and reducing database load by 20%
- Resolved a critical bug in the self-service SAML workflow preventing customers from logging in after configuring SAML SSO, by debugging complex authentication code and implementing a fix

SAP — *Jan 2022 - Sept 2022*
Software Engineer Intern — *Vancouver, BC*

- Designed a Kubernetes-based synthetic monitoring service for health checks on critical workflows using a pub/sub architecture with RabbitMQ, improving product reliability
- Minimized customer impact by proactively alerting on-call developers of issues by creating alert systems for key metrics such as error rate, **saved 20% of their time**

Project Experience

GameBoy Emulator
Rust

- Built a GameBoy emulator in Rust, implementing core hardware components such as the CPU, memory bus, and PPU
- Emulated hardware behavior from specifications, including interrupts, with cycle-accurate timing for hardware synchronization

NP-Compete
TypeScript, Node.js, Express, Redis, TypeORM, PostgreSQL, Next.js, Chakra UI

- Developed a web application where users join rooms to compete in programming challenges against each other
- Created REST APIs in Express providing endpoints allowing users to login, create rooms, and submit code
- Implemented rooms using WebSockets to communicate events between the clients and server in real-time
- Utilized Redis to persist user sessions and cache frequently accessed data, improving application performance

URL Shortener
Go, gRPC, Docker, Kubernetes, Redis, Cassandra, Amazon Web Services (AWS)

- Created a scalable URL shortener in Go, employing a microservice architecture with efficient gRPC communication
- Deployed on AWS using EKS for Kubernetes, leveraging automated scaling, load balancing, and self-healing
- Integrated Redis for high performance caching, enhancing read performance and reducing database load
- Used Cassandra for data replication and partitioning, ensured high availability and scalability with large data volumes

Skills

Languages: TypeScript, JavaScript, Java, Java, Rust, C, C++, Go, SQL, x86 Assembly, HTML, CSS
Technologies: Node.js, Express, Spring Boot, GraphQL, React, Next.js, Redis, PostgreSQL, OpenID Connect, SAML, OAuth
Tools: Git, GitHub, Jira, Jenkins, Docker, Kubernetes, Grafana, JMeter, Playwright, Claude Code, GitHub Copilot